

Strawbees Kit

WHAT'S IN THE KIT

- Strawbees connectors and construction straws



QUICK-START GUIDE

1. Connectors slide into straw ends — demo the pinch-and-push.
2. Start with 2D shapes, then combine into 3D structures.
3. Triangles are strong, squares are wobbly — let students discover this.
4. Cut straws to custom lengths for advanced builds.

FULL LESSONS IN THIS GUIDE

GRADES K-2	Shape Detectives	page 2
GRADES 3-5	Geodesic Dome Build	page 3
GRADES 6-8	Moving Mechanisms	page 4

PRO TIP: Nearly silent and mess-free — one of the best kits for classrooms with tight cleanup windows.

Strawbees Kit

LESSON · GRADES K-2

Shape Detectives

LEARNING OUTCOMES

- Students will construct 2D shapes and identify them by name
- Students will discover that triangles resist deformation
- Students will find structural shapes in real buildings

MATERIALS

- Strawbees kit (straws + connectors)
- Shape cards (triangle, square, pentagon)
- Photos of bridges, towers, and roofs
- Squish-test protocol: gentle press from the top

INSTRUCTIONS

1. Teach the pinch-and-push connection with one straw and one connector.
2. Groups build each shape card in straws.
3. Squish test: press each shape gently — which ones hold, which collapse?
4. Sort shapes into 'strong' and 'squishy' piles based on evidence.
5. Detective hunt: find and circle the triangles in building photos.
6. Close: why do builders hide triangles inside everything?

ACTIVITY

Shape Detectives — build shapes, squish them, and discover the triangle's secret, then find it hiding in every bridge photo. Geometry meets structural intuition at age six.

Strawbees Kit

LESSON · GRADES 3-5

Geodesic Dome Build

LEARNING OUTCOMES

- Students will construct a 3D geodesic structure from repeated triangles
- Students will test load-bearing capacity fairly
- Students will connect shape repetition to strength

MATERIALS

- Strawbees kit
- Dome assembly guide (or projected build steps)
- Standard load weights (identical books)
- Team role cards: connector, counter, checker

INSTRUCTIONS

1. Warm-up: build and squish-test a triangle vs. a square as review.
2. Assign team roles; walk through dome construction ring by ring.
3. Domes complete when they stand unsupported.
4. Load test: add books one at a time on a cardboard platform; record the max.
5. Compare domes: did tighter connections or better symmetry hold more?
6. Extension: repair and reinforce, then retest for a second attempt.

ACTIVITY

Geodesic Dome Build — many triangles, one surprisingly strong dome. The load test with identical books turns 'it looks strong' into a measured claim.

Strawbees Kit

LESSON · GRADES 6-8

Moving Mechanisms

LEARNING OUTCOMES

- Students will design a linkage mechanism that transfers motion
- Students will prototype and iterate joints rapidly
- Students will analyze degrees of freedom informally

MATERIALS

- Strawbees kit
- Mechanism challenge cards: grabber claw, extending arm, drawing machine
- Target objects to grab or reach
- Markers and paper for drawing-machine option

INSTRUCTIONS

1. Demo one working scissor linkage — extend and collapse it slowly.
2. Teams pick a challenge card and sketch their mechanism first.
3. Prototype the core joint before building the full mechanism.
4. Iterate: floppy joints and binding joints both demand redesign.
5. Function test against the card's criteria (grab it, reach it, draw with it).
6. Mechanism show-and-tell focused on the joint that took the most tries.

ACTIVITY

Moving Mechanisms — build grabbers, extenders, and drawing machines with near-instant iteration. Straw joints fail fast and cheap, which is exactly what makes them great teachers.